
Final Group Project: Deep Learning for Regulatory DNA Sequence Optimization

1. Introduction

In your individual projects, you built models to model the relationship between DNA sequences and regulatory activity. For this final group project, you will transition from **evaluating** sequences to **engineering** them, using your deep learning models as a "simulation environment" to either repair or engineer synthetic DNA sequences.

You will be evaluated on your ability to use **model interpretation techniques** to guide your decisions and your capacity to defend these choices in your presentation.

2. Subtask A: DNA Restoration Challenge

In this task, you will receive **3 DNA sequences** originally derived from high-activity enhancers. These sequences have been intentionally disrupted by introducing mutations that reduced their regulatory function.

Unlike a binary classification setting, these sequences remain partially functional, but exhibit significantly reduced activity scores compared to their original form.

- **Objective:** Recover and improve the regulatory activity of each sequence.
- **The Goal:** Maximize the predicted `rna_dna_ratio` while recovering or exceeding the original enhancer activity signal.
- **Constraint:** You are limited to the amount of mutations that were introduced

Details about the sequences

Each sequence originates from a high-activity enhancer but has been independently perturbed:

Sequence	Original Activity (approx.)	Introduced Mutations
Seq 1	~2.5	40 mutations
Seq 2	~2.7	16 mutations
Seq 3	~1.1	20 mutations

3. Subtask B: The Super-Enhancer Challenge

You will be given **1 sequence** with naturally low regulatory activity. For this task, there are no limits on how much you can modify the DNA.

- **Objective:** Re-engineer this sequence to achieve the **absolute maximum activity score** possible according to your models.

- **No mutation budget.** You may modify as many nucleotides as needed.
- **The Goal:** Design an active sequence that outperforms the sequences found in the original training set.
- **The Constraint:** The original input sequence must be preserved as the starting point, and your approach should demonstrate iterative refinement guided by model feedback.

Note on Biological Realism: While models can sometimes be "tricked" by simple repetitive patterns (like GC repeats), a robust model should reward complex **biological motifs**. Your designs will be examined for biological plausibility during the defense.

4. Deep Learning Methodology

Your approach must be **model-guided rather than based on random trial and error**. You can consider the following strategies, but you are free to use any methods you know:

- **Gradient Attribution:** Use backpropagation to see which bases your model is most sensitive to.
- **In Silico Scanning:** Write scripts to test possible mutations and observe the predicted change in the `rna_dna_ratio`.
- **Ensemble Consensus:** Validate your designs against **all three** group members' models to ensure the features you created are robust and not just overfitting to one specific architecture.

5. Presentation & Defense

The final evaluation consists of a **15-minute presentation** and a Q&A session. You must describe your architectural choices.

Required Presentation Elements:

- **The Optimization Curve:** For the "Engineered-Enhancer", show a plot of the predicted activity climbing as you modified the sequence.
- **Model Conflict:** Discuss any moments where your group's models disagreed on a mutation and how you resolved them.
- **Motif Identification (Optional):** Identify whether you recreated real biological motifs (e.g., TATA-boxes or specific transcription factor binding sites).

6. Grading (5 Points Total)

- **Technical Execution (2.0 Points):** Success in restoring "Broken" sequences and the peak activity achieved in the "Engineered-Enhancer" task.
- **Deep Learning Analysis (1.5 Points):** Effective use of attribution methods to guide and justify mutations.
- **Presentation & Defense (1.5 Points):** Clarity of explanation, quality of visualizations, and the ability to defend your approach.

Final sequence activity will be benchmarked using an external reference model not accessible during development, encouraging robust and generalizable optimization strategies.

7. Deliverables

- **Optimized Sequence File:** A TSV with `id`, `new_sequence`, and your group's `predicted_rna_dna_ratio`.
- **The Solution Notebook:** A documented notebook/script showing your solution.
- **Presentation Slides:** Including all required visualizations.